Crash Reporting

Data Analytics w/ Python

Group 5
Maheshwari Bhandare
Shwetalee Gadekar
Divya Parekh
Ankita Dhekane

This dataset captures detailed road crash information, including crash date, location, road conditions, vehicle details, driver behavior, and environmental factors. It's designed to help analyze crash causes and outcomes.

Key Columns

- **Crash Date/Time** - Date and Time of crash.
- **Collision Type** - Type of collision.
- **Driver At Fault** - Whether this driver was at fault.
- **Injury Severity** - Severity of injury to this driver.
- **Vehicle Movement** - The movement of the vehicle at the time of the collision.
- **Traffic Control** - Signage or traffic control devices.
- **Weather** - Weather at collision location.
- **Surface Condition** - Condition of roadway surface.
- **Light** - Lighting conditions.
- **Driver Substance Abuse** - Substance abuse detected for all drivers involved.
- **Driver Distracted By** - The reason the driver was distracted.
- **Latitude** - Y coordinate of crash location.
- **Longitude** - X coordinate of crash location.

Data Preparation & Cleaning

Data Preparation and Cleaning

1. Dropped columns with more than 80% of NaN value on average

```
Index(['Off-Road Description', 'Municipality', 'Related Non-Motorist',
       'Non-Motorist Substance Abuse', 'Circumstance'],
      dtype='object')
```

Considered Columns

```
Index(['Local Case Number', 'Agency Name', 'ACRS Report Type',
       'Crash Date/Time', 'Route Type', 'Road Name', 'Cross-Street Name',
       'Collision Type', 'Weather', 'Surface Condition', 'Light',
       'Traffic Control', 'Driver Substance Abuse', 'Person ID',
       'Driver At Fault', 'Injury Severity', 'Driver Distracted By',
       'Drivers License State', 'Vehicle ID', 'Vehicle Damage Extent',
       'Vehicle First Impact Location', 'Vehicle Body Type',
       'Vehicle Movement', 'Vehicle Going Dir', 'Speed Limit',
       'Driverless Vehicle', 'Parked Vehicle', 'Vehicle Year', 'Vehicle Make',
       'Vehicle Model', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
```

Data Preparation and Cleaning

2. Replaced missing and 'OTHER' values with 'UNKNOWN' for uniformity

Before

	Crash Date/Time	Vehicle Movement	Road Name	Agency Name	Weather	Collision Type	Injury Severity	Vehicle Movement
172062	05/13/2024 10:30:00 PM	Turning Left	ODENDHAL AVE (EB/L)	GAITHERSBURG	Clear	Other	Suspected Minor Injury	Turning Left
172070	05/05/2024 12:18:00 PM	Slowing or Stopping	FLORAL ST	MONTGOMERY	Unknown	Other	NaN	Slowing or Stopping

After

	Crash Date/Time	Vehicle Movement	Road Name	Agency Name	Weather	Collision Type	Injury Severity	Vehicle Movement
172062	2024-05-13 22:30:00	TURNING LEFT	ODENDHAL AVE (EB/L)	GAITHERSBURG	CLEAR	UNKNOWN	SUSPECTED MINOR INJURY	TURNING LEFT
172124	2024-05-13 22:30:00	MOVING CONSTANT SPEED	ODENDHAL AVE (EB/L)	GAITHERSBURG	CLEAR	UNKNOWN	SUSPECTED MINOR INJURY	MOVING CONSTANT SPEED

Data Preparation and Cleaning

3. Standardize text data - Converted all string columns to uppercase for consistency

Before

	Crash Date/Time	Vehicle Movement	Agency Name	Weather	Collision Type	Injury Severity	Vehicle Movement
172130	04/25/2024 09:00:00 PM	Moving Constant Speed	MONTGOMERY	Clear	Unknown	No Apparent Injury	Moving Constant Speed
172500	02/16/2024 07:33:00 PM	Moving Constant Speed	MONTGOMERY	Clear	Unknown	No Apparent Injury	Moving Constant Speed

After

	Crash Date/Time	Vehicle Movement	Agency Name	Weather	Collision Type	Injury Severity	Vehicle Movement
0	2021-05-27 19:40:00	PARKED	TAKOMA PARK POLICE DEPART	CLEAR	UNKNOWN	NO APPARENT INJURY	PARKED
1	2015-09-11 13:29:00	PARKING	MONTGOMERY	CLEAR	UNKNOWN	NO APPARENT INJURY	PARKING

Data Preparation and Cleaning

4. Column-wise cleaning for unique values (5 columns)

Before

```
array([nan, 'DRY', 'ICE', 'WET', 'SLUSH', 'UNKNOWN',
       'WATER(STANDING/MOVING)', 'SNOW', 'OTHER', 'MUD, DIRT, GRAVEL',
       'OIL', 'SAND', 'Dry', 'Wet', 'Other', 'Water (standing, moving)',
       'Ice/Frost', 'Mud, Dirt, Gravel', 'Snow', 'Slush', 'Sand'],
      dtype=object)
```

After

```
array(['UNKNOWN', 'DRY', 'ICE', 'WET', 'SLUSH', 'SNOW',
       'MUD, DIRT, GRAVEL', 'OIL', 'SAND'], dtype=object)
```

Data Preparation and Cleaning

5. Crash Date/Time column converted to 24-hour format

Before

	Crash Date/Time
172062	05/13/2024 10:30:00 PM
172070	05/05/2024 12:18:00 PM

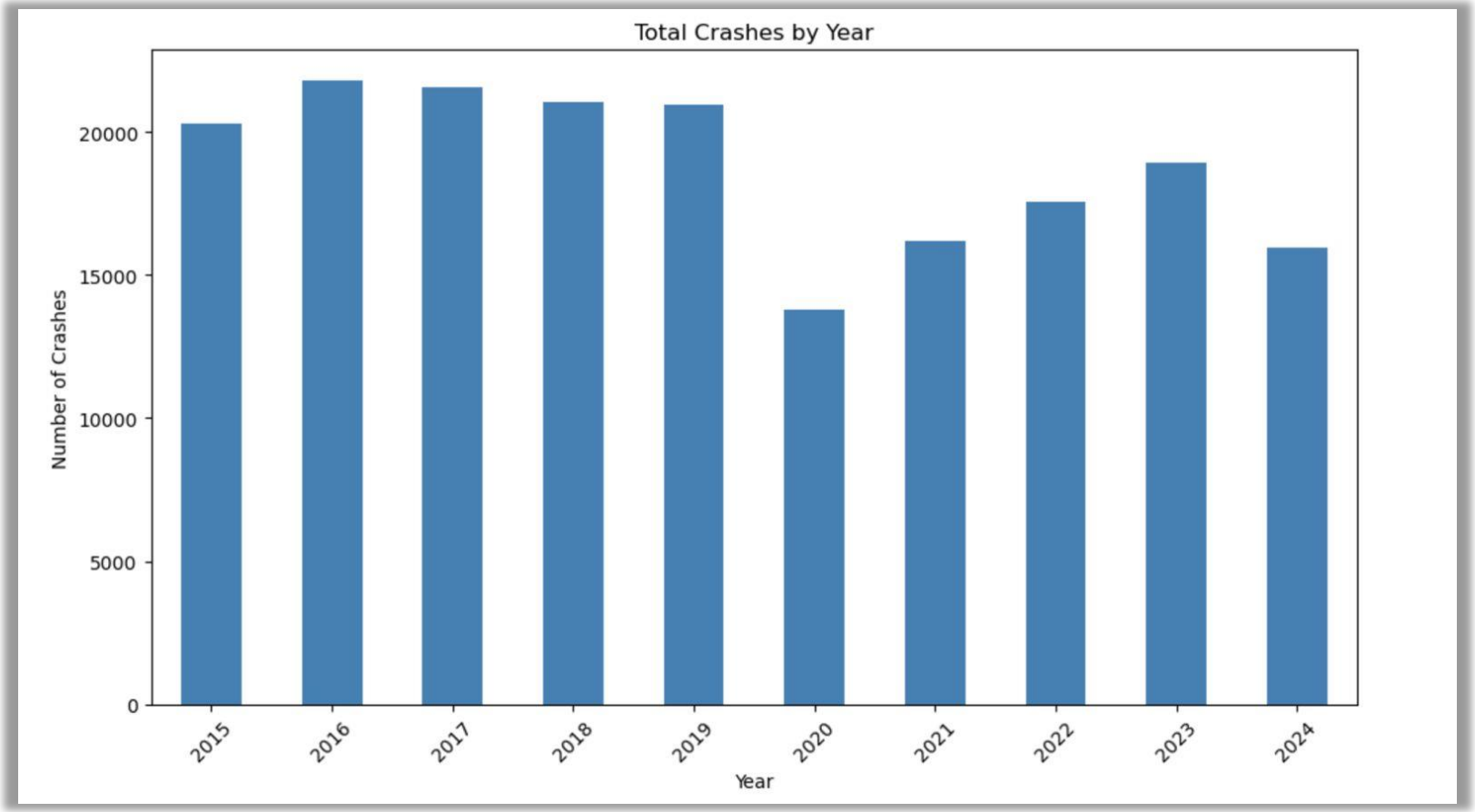
After

	Crash Date/Time
172062	2024-05-13 22:30:00
172124	2024-05-13 22:30:00

Data Exploration

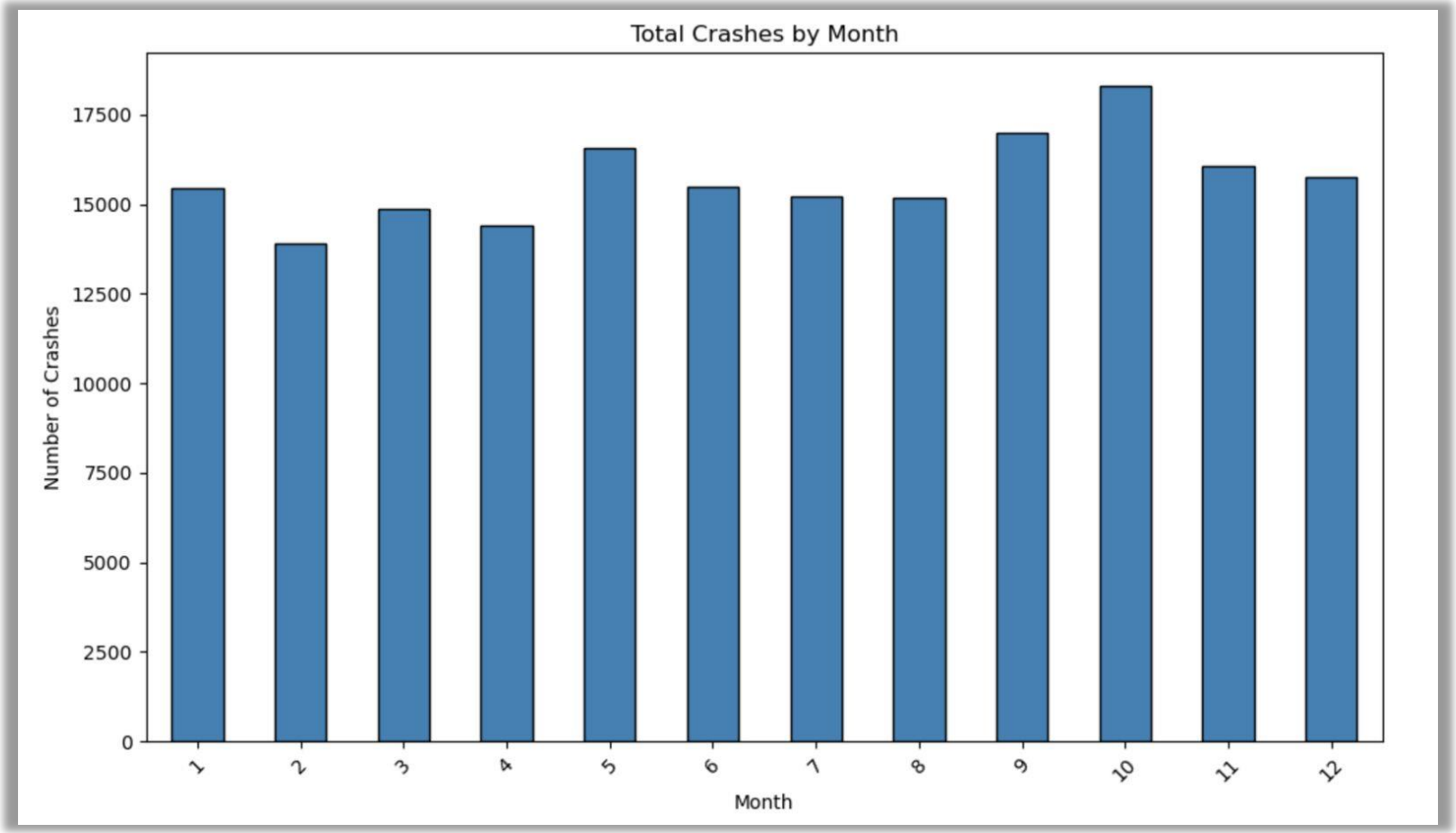
Data Exploration

The chart shows how events like the pandemic can have a significant impact on road activity, with crashes decreasing during lockdowns, followed by the heavy adoption of WFH, and gradually increasing as normal life resumes.



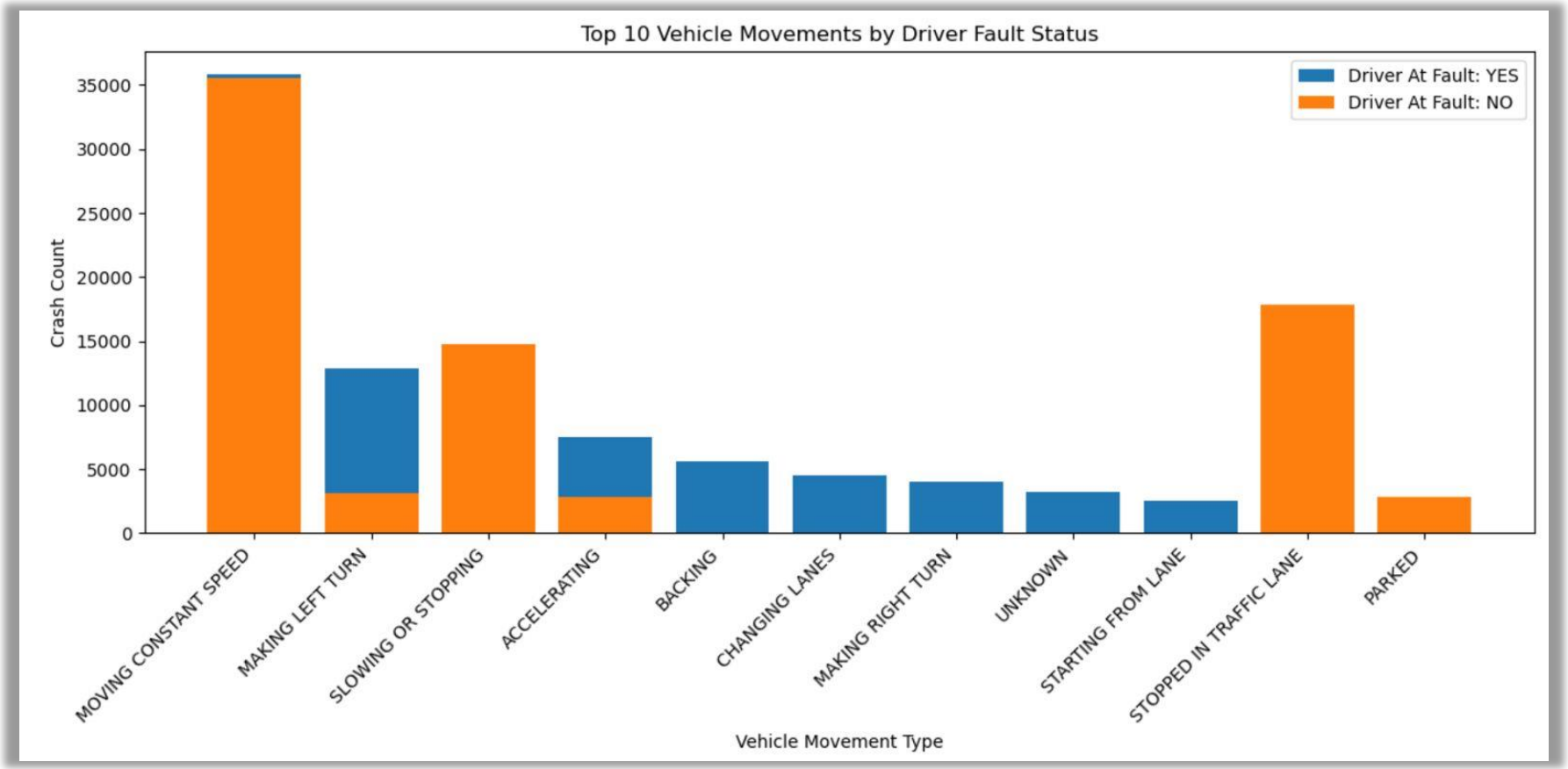
Data Exploration

Crashes are relatively consistent across the year. Slightly higher crashes occur in September and October due to seasonal changes, back-to-school traffic, festivals and celebrations like Halloween.



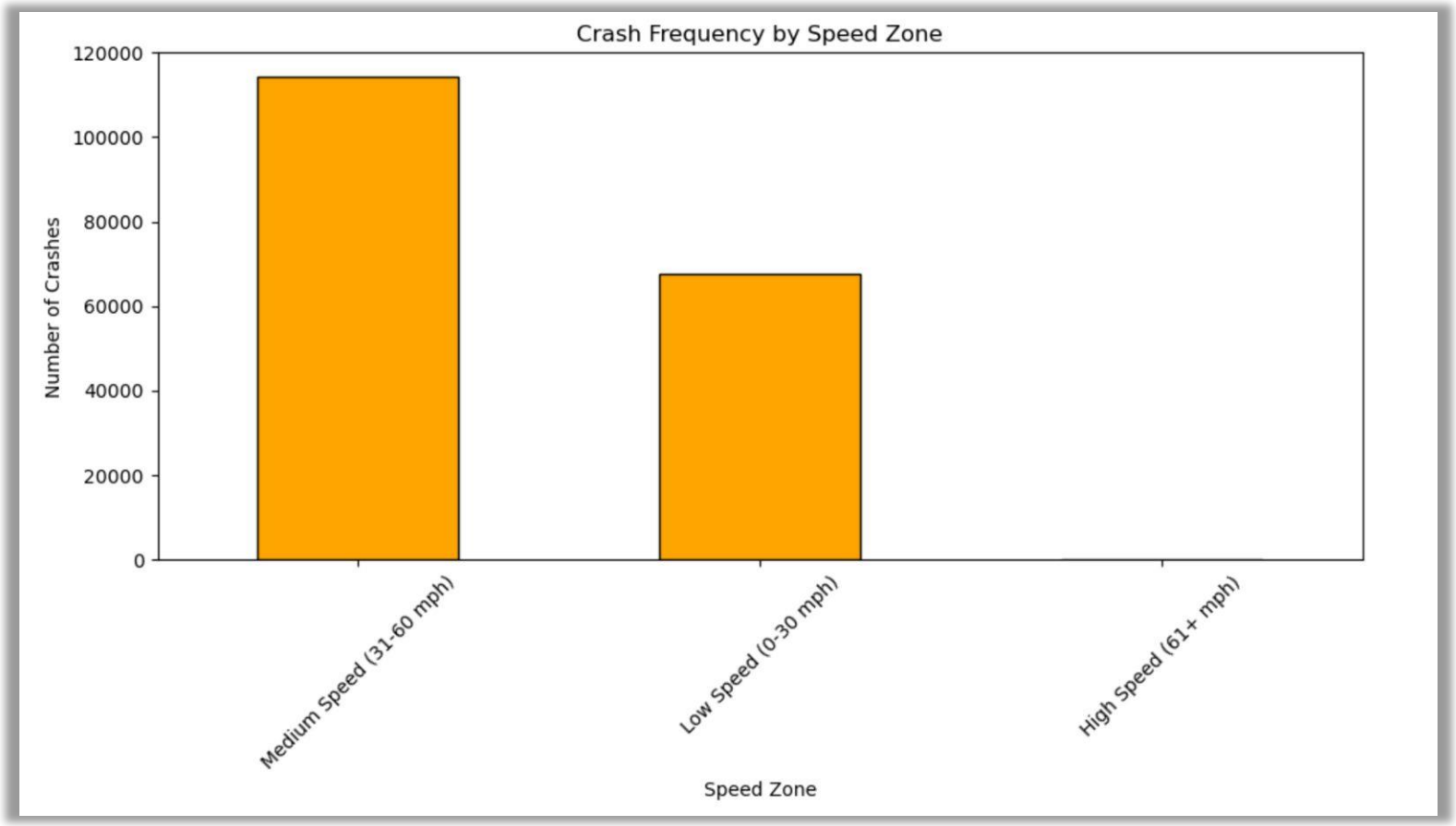
Data Exploration

This chart helps identify high-risk vehicle movements and their correlation with driver fault, providing valuable insights for road safety strategies.



Data Exploration

This analysis highlights the relationship between speed zones and crash frequency, emphasizing the need for targeted safety measures in medium and low-speed zones.

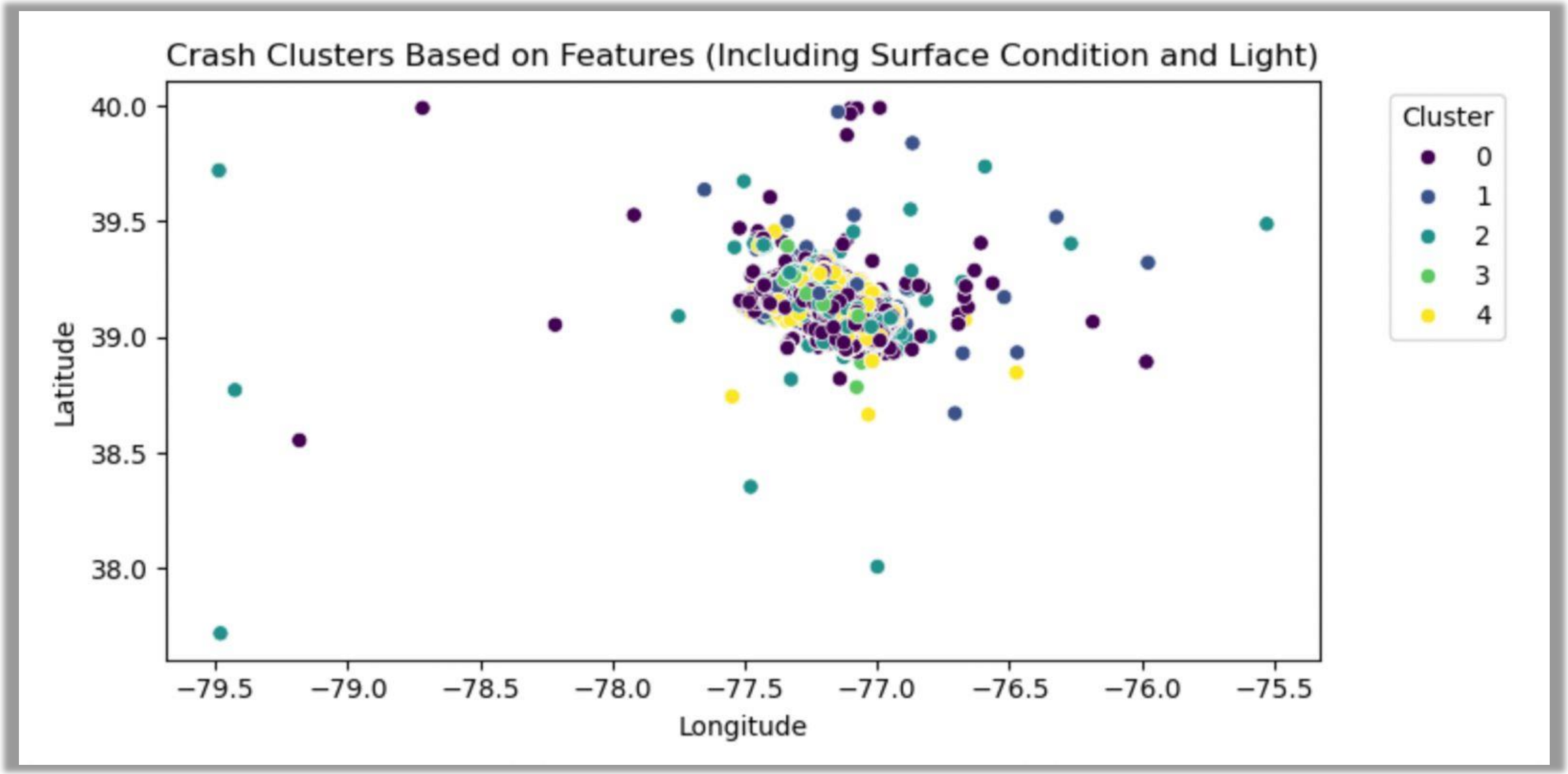


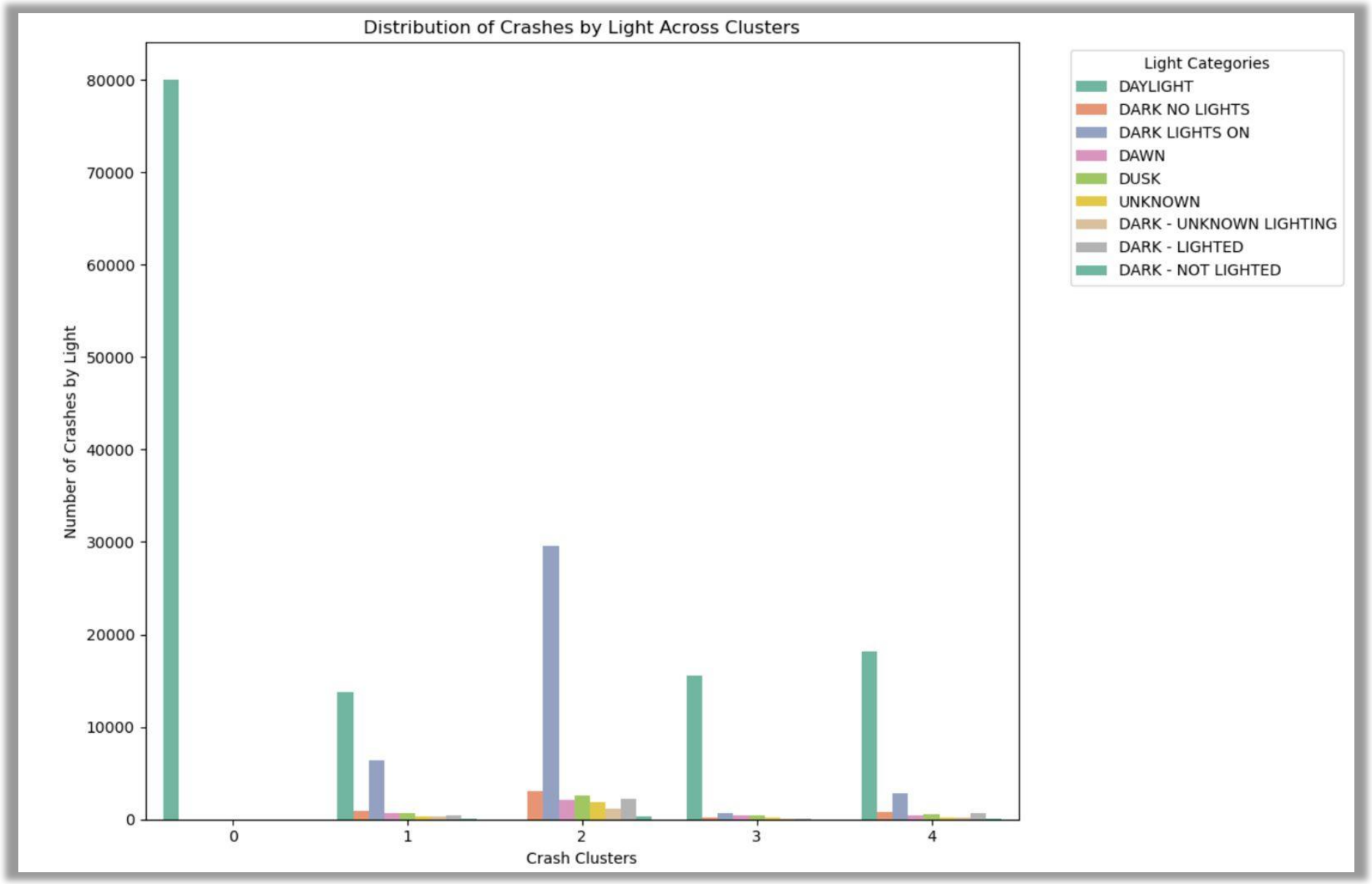
Findings

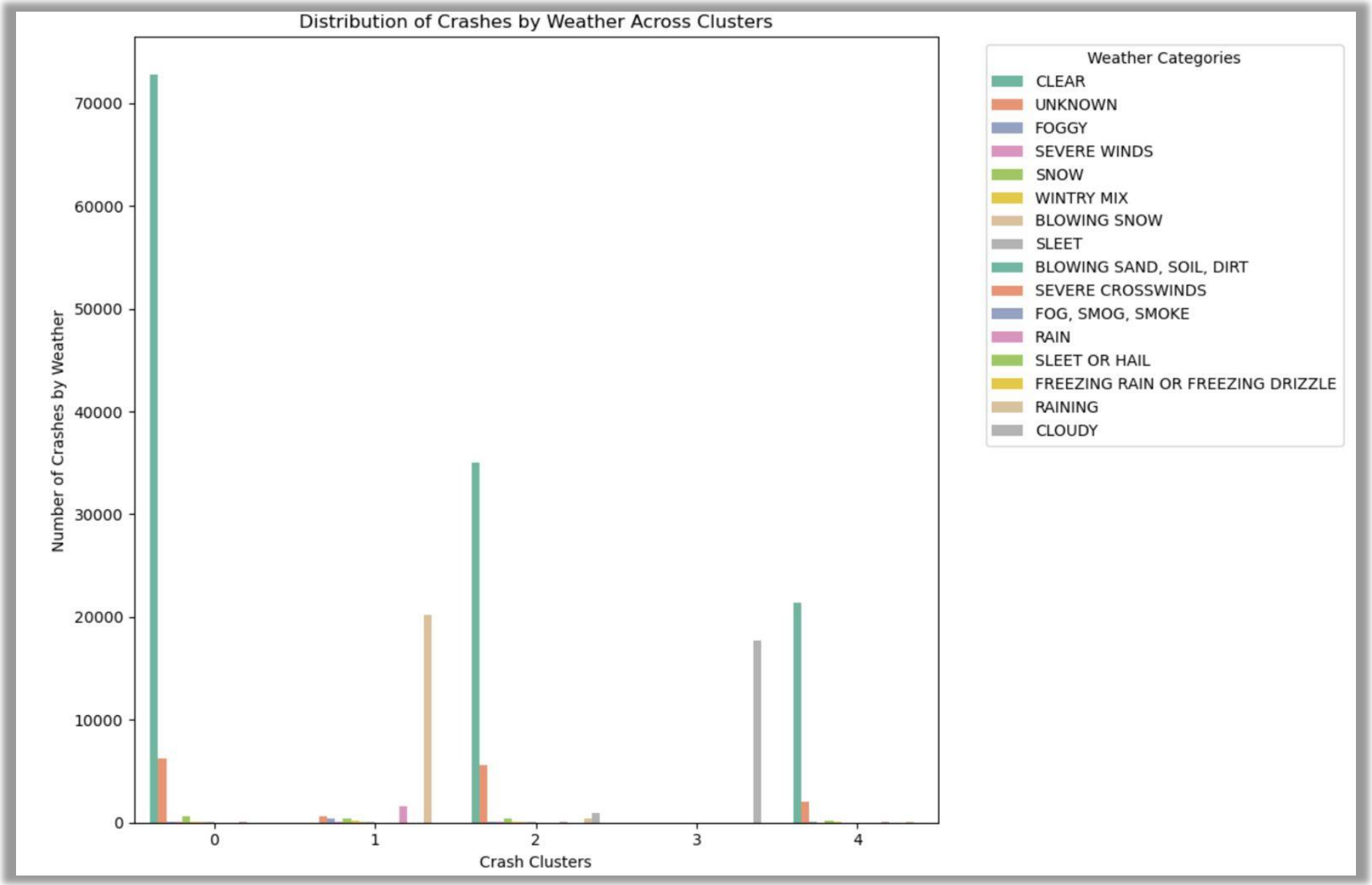
Finding #1: Weather Impact on Crash Incidents

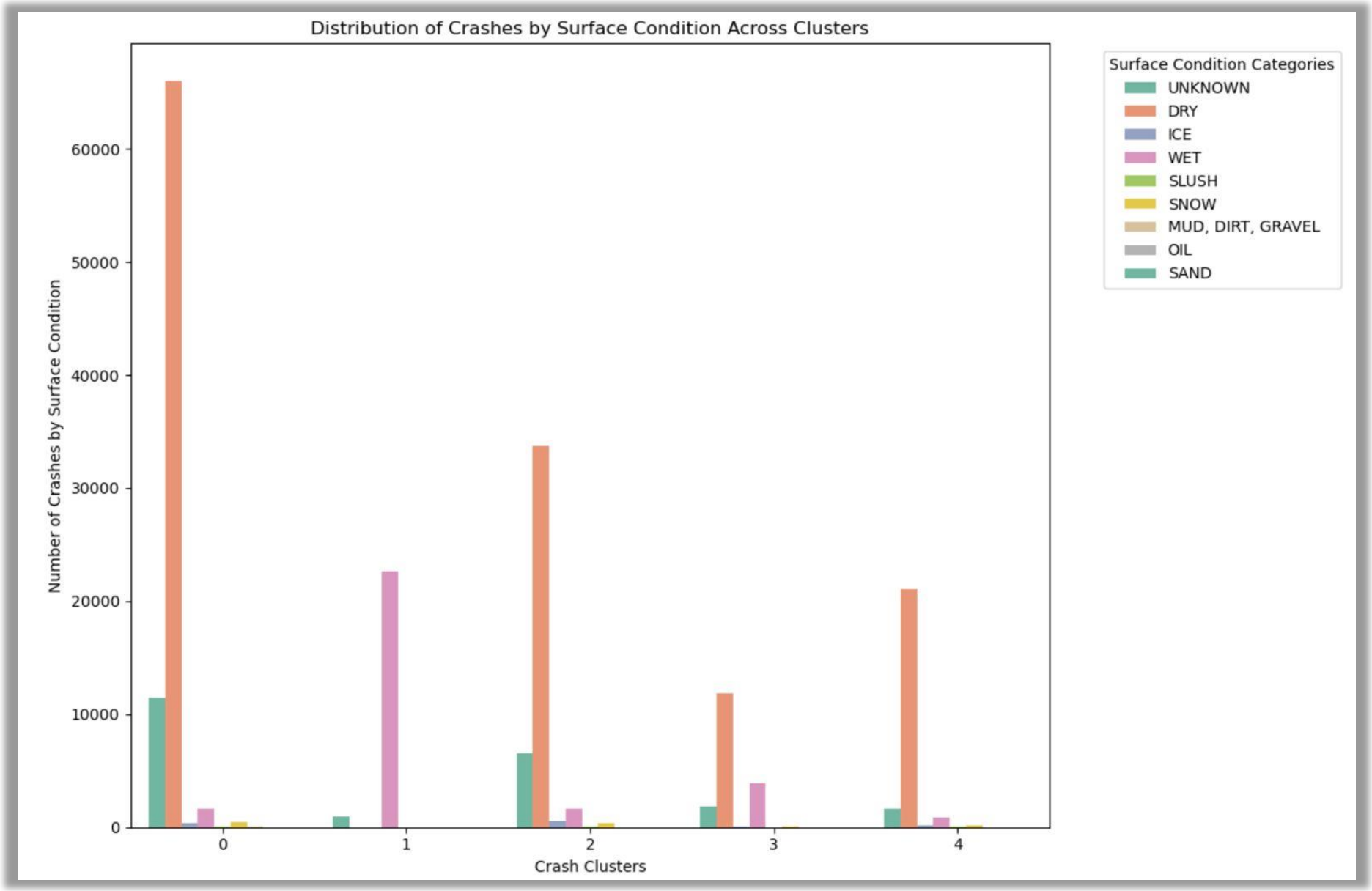
Crashes peaked under conditions often deemed safest—clear weather, dry surfaces, and daylight—highlighting that optimal environments alone cannot prevent accidents.

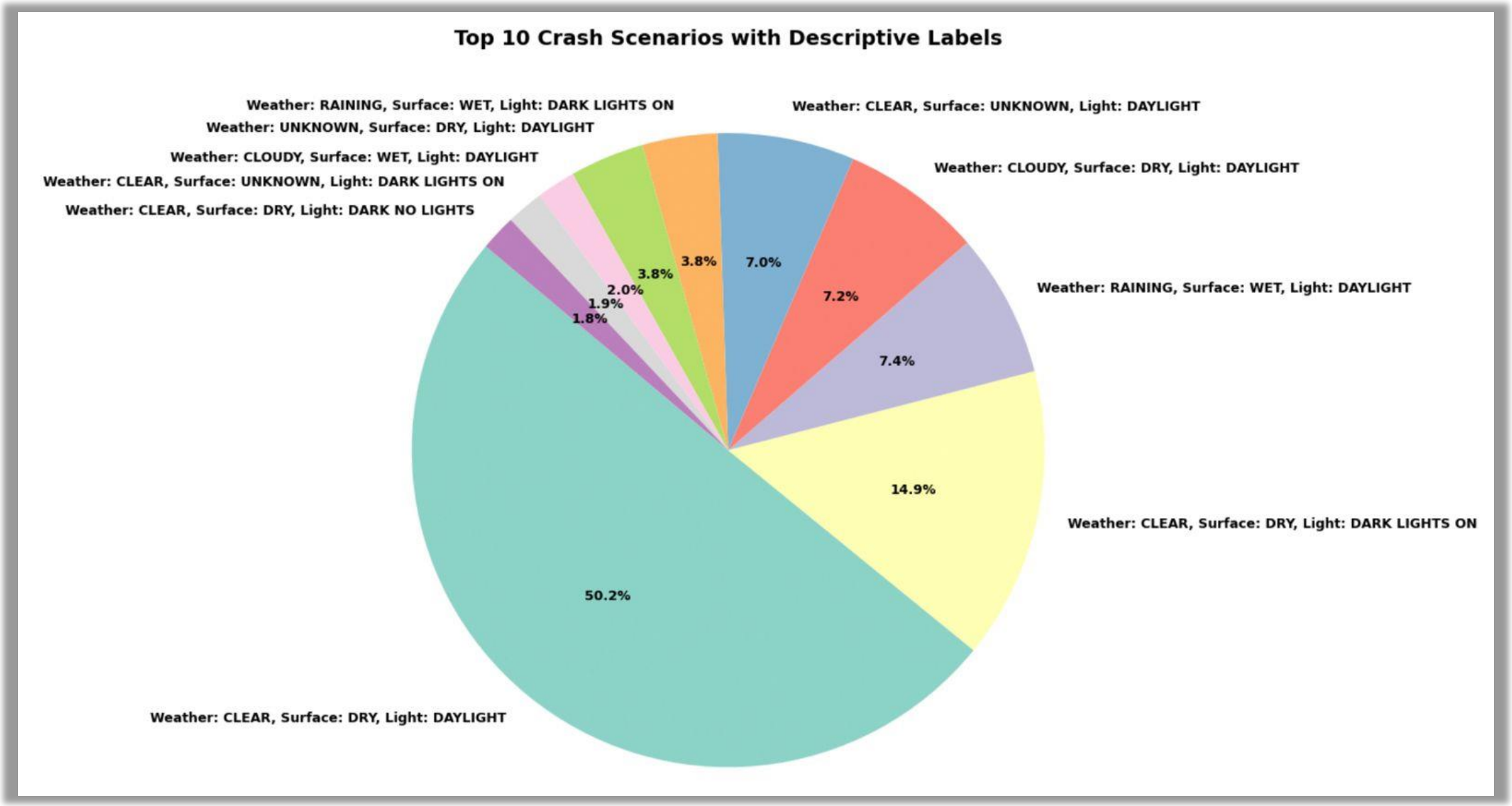
Clustering applied to identify accident patterns under varied environmental conditions


Crash Clusters Based on Features (Including Surface Condition and Light)
Latitude
40.0
39.5
39.0
38.5
38.0
−79.5
−79.0
−78.5
−78.0
−77.5
−77.0
−76.5
−76.0
−75.5
Longitude
Cluster
0
1
2
3
4


Distribution of Crashes by Light Across Clusters
Number of Crashes by Light
80000
70000
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
Crash Clusters
Light Categories
DAYLIGHT
DARK NO LIGHTS
DARK LIGHTS ON
DAWN
DUSK
UNKNOWN
DARK - UNKNOWN LIGHTING
DARK - LIGHTED
DARK - NOT LIGHTED

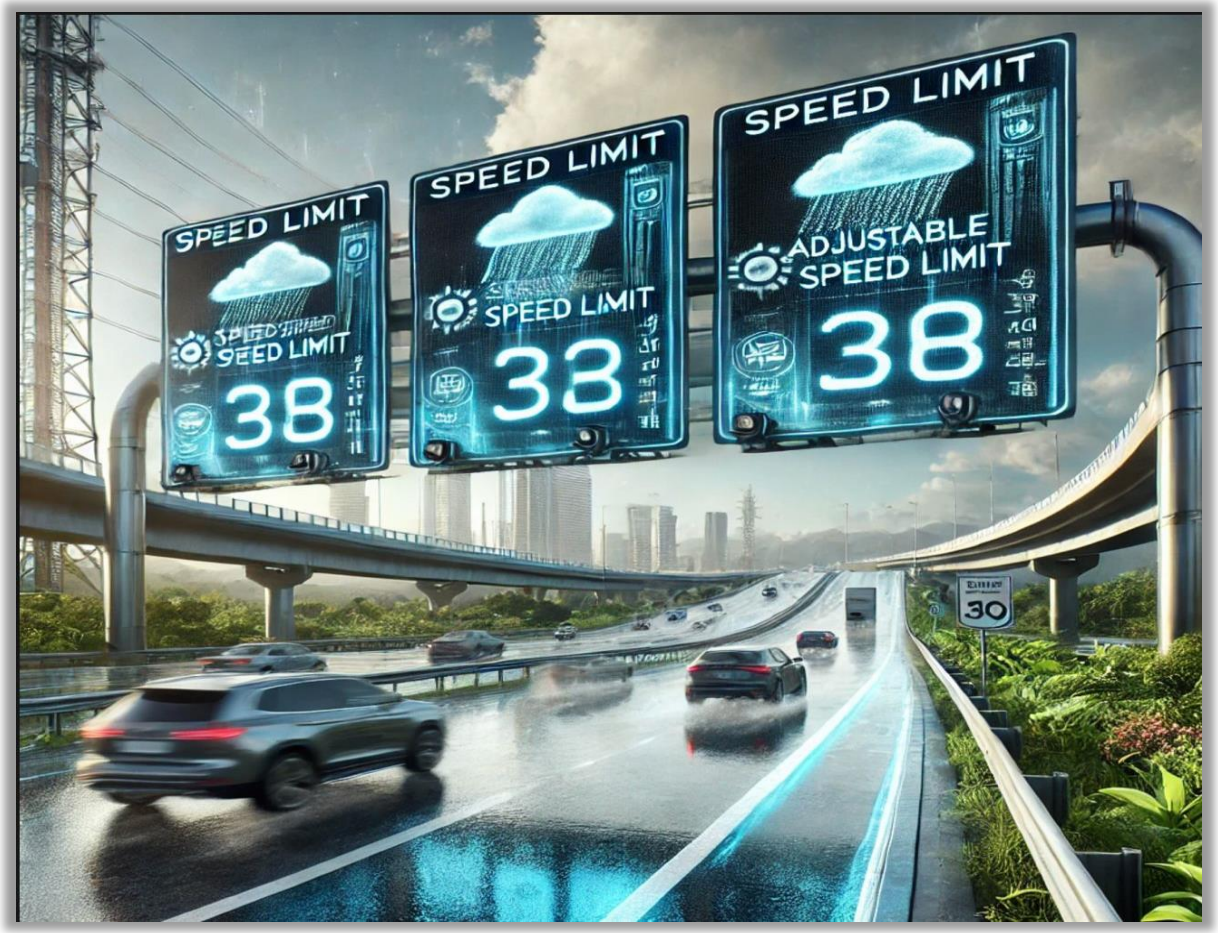

Distribution of Crashes by Weather Across Clusters
Number of Crashes by Weather
70000
60000
50000
40000
30000
20000
10000
0
0
1
2
3
4
Crash Clusters
Weather Categories
CLEAR
UNKNOWN
FOGGY
SEVERE WINDS
SNOW
WINTRY MIX
BLOWING SNOW
SLEET
BLOWING SAND, SOIL, DIRT
SEVERE CROSSWINDS
FOG, SMOG, SMOKE
RAIN
SLEET OR HAIL
FREEZING RAIN OR FREEZING DRIZZLE
RAINING
CLOUDY


Distribution of Crashes by Surface Condition Across Clusters
Number of Crashes by Surface Condition
0
10000
20000
30000
40000
50000
60000
0
1
2
3
4
Crash Clusters
Surface Condition Categories
UNKNOWN
DRY
ICE
WET
SLUSH
SNOW
MUD, DIRT, GRAVEL
OIL
SAND


Top 10 Crash Scenarios with Descriptive Labels
Weather: RAINING, Surface: WET, Light: DARK LIGHTS ON
Weather: UNKNOWN, Surface: DRY, Light: DAYLIGHT
Weather: CLOUDY, Surface: WET, Light: DAYLIGHT
Weather: CLEAR, Surface: UNKNOWN, Light: DARK LIGHTS ON
Weather: CLEAR, Surface: DRY, Light: DARK NO LIGHTS
Weather: CLEAR, Surface: UNKNOWN, Light: DAYLIGHT
Weather: CLOUDY, Surface: DRY, Light: DAYLIGHT
Weather: RAINING, Surface: WET, Light: DAYLIGHT
Weather: CLEAR, Surface: DRY, Light: DARK LIGHTS ON
Weather: CLEAR, Surface: DRY, Light: DAYLIGHT
7.0%
7.2%
7.4%
14.9%
50.2%
3.8%
3.8%
2.0%
1.9%
1.8%

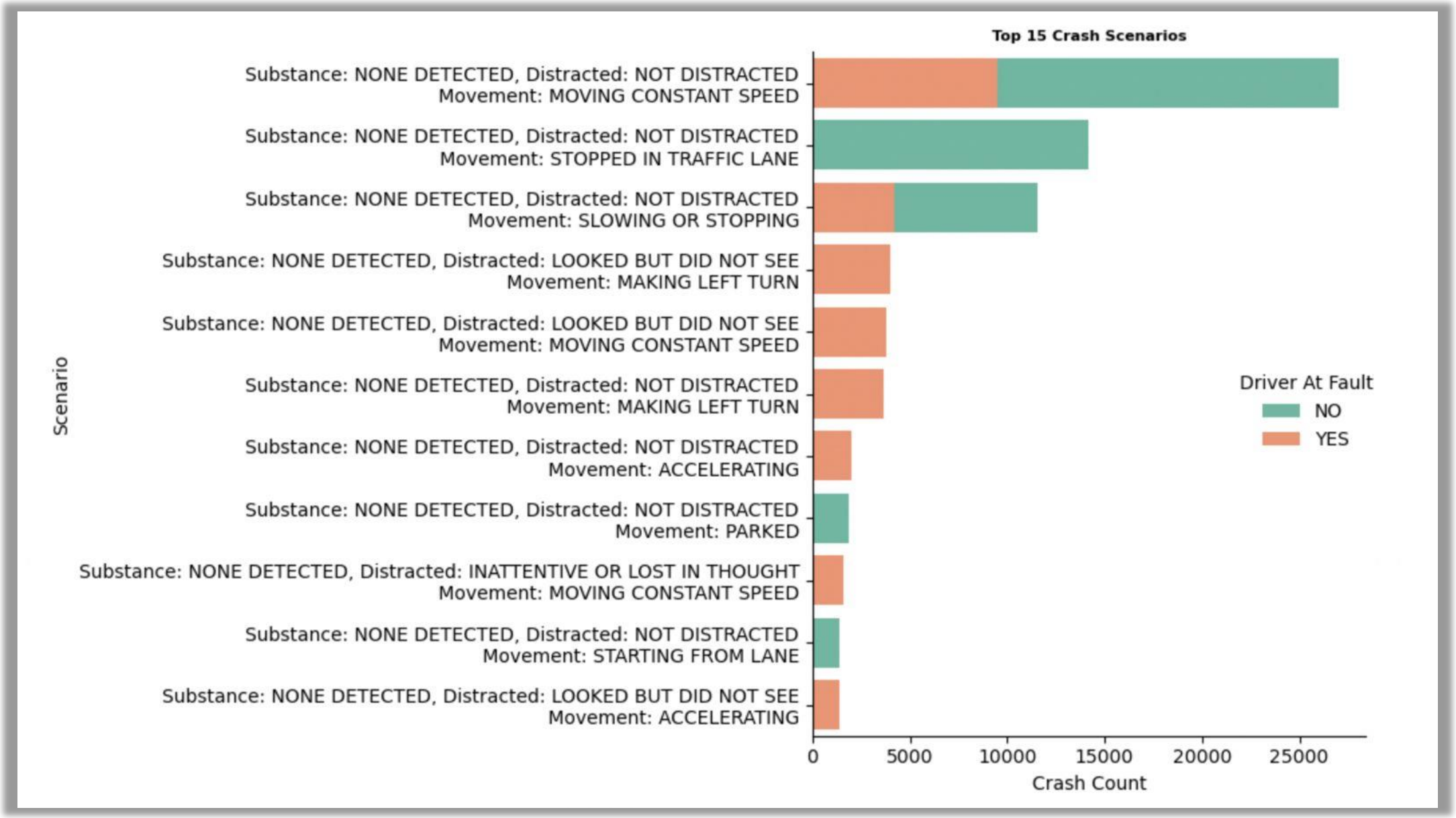
Managerial Insights

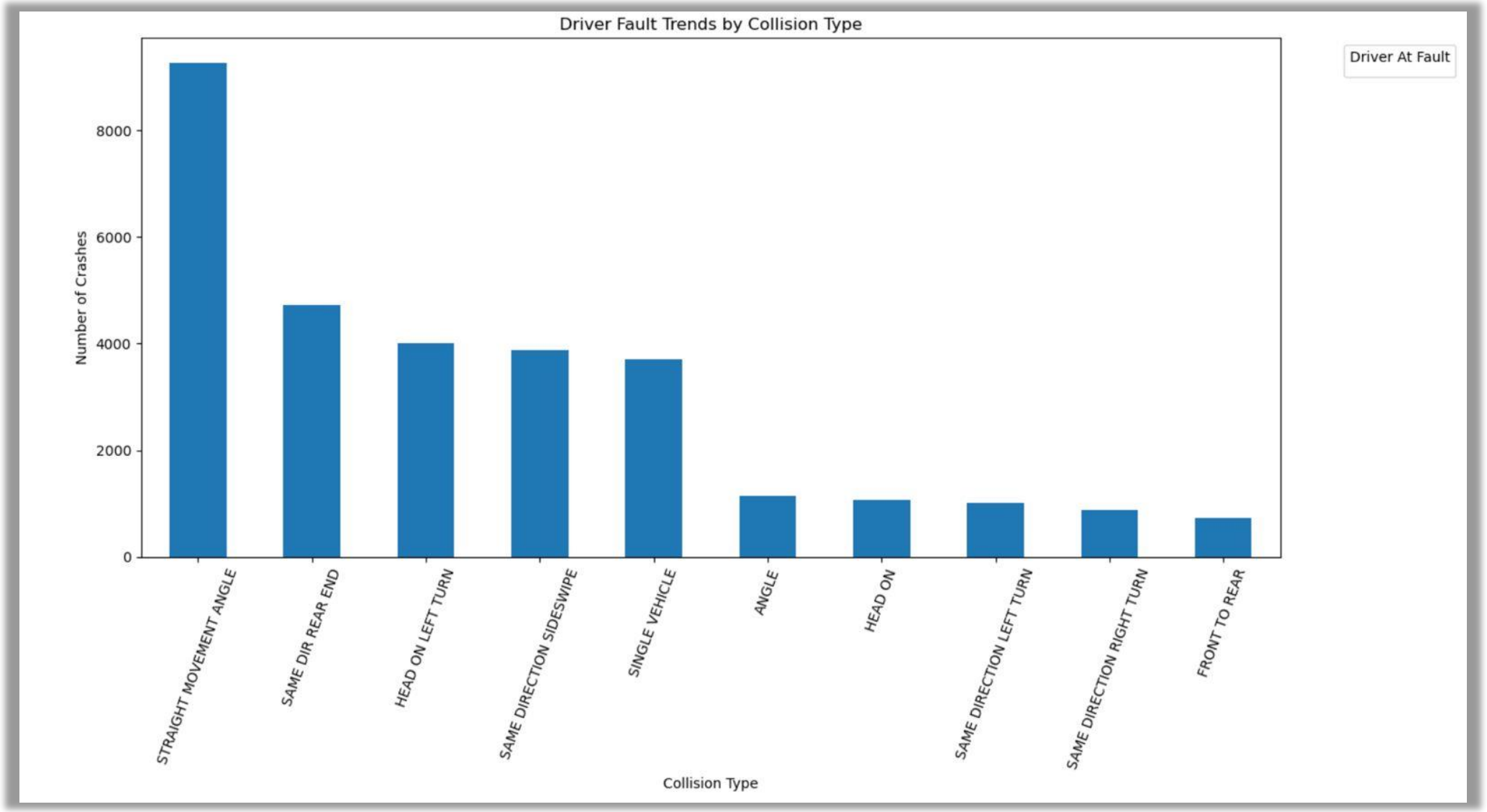
Use adjustable speed limits that change based on traffic or weather conditions. For example, speed limits could lower automatically if traffic is heavy or visibility is poor.

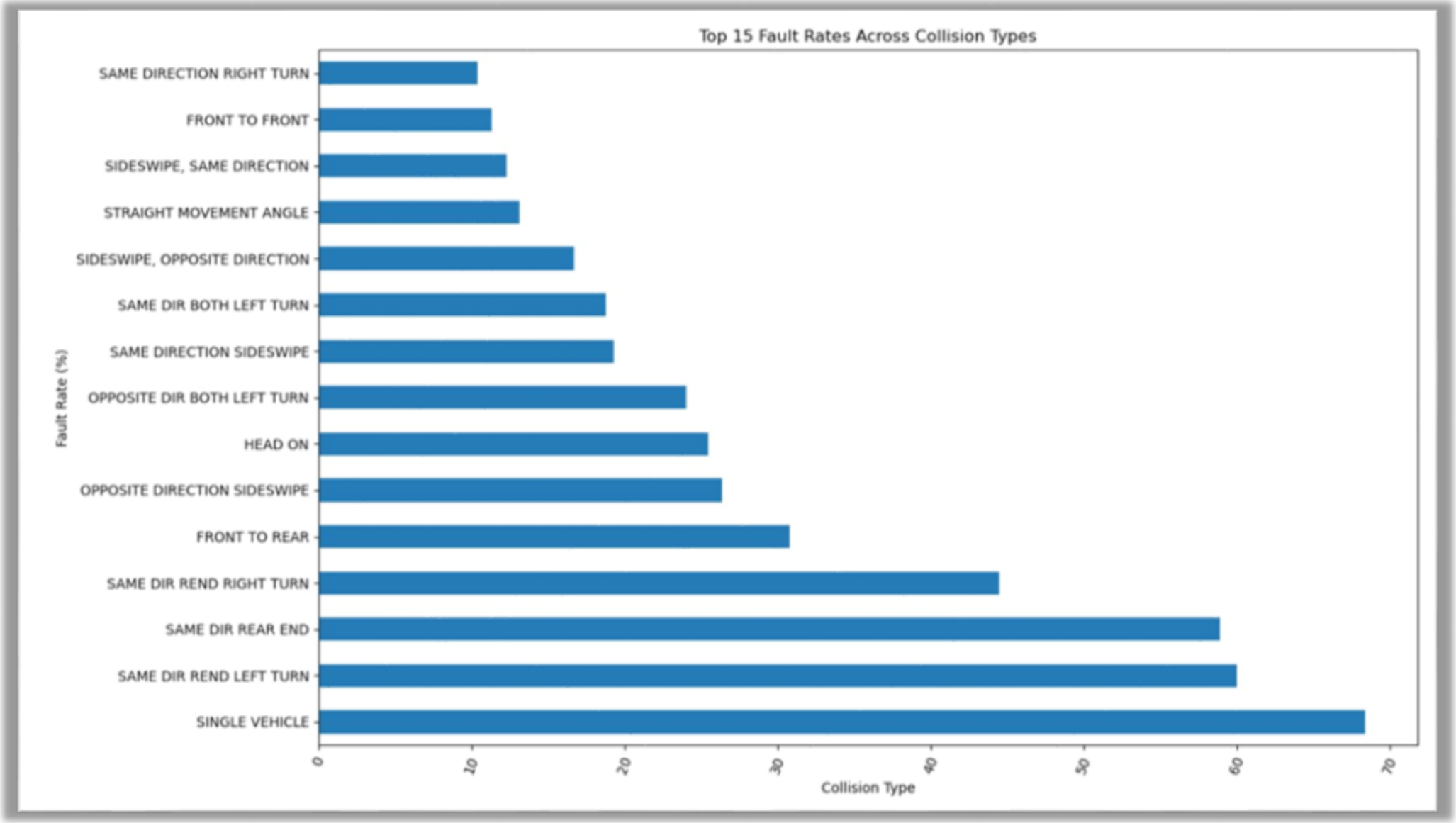


Finding #2: Higher Driver’s Fault rates found in Solo Crashes

Collision types involving individual vehicle control (e.g., single vehicle, rear-end movement) exhibit higher fault rates. Situations with shared driver responsibilities (e.g., sideswipes, turns) tend to have lower fault rates.

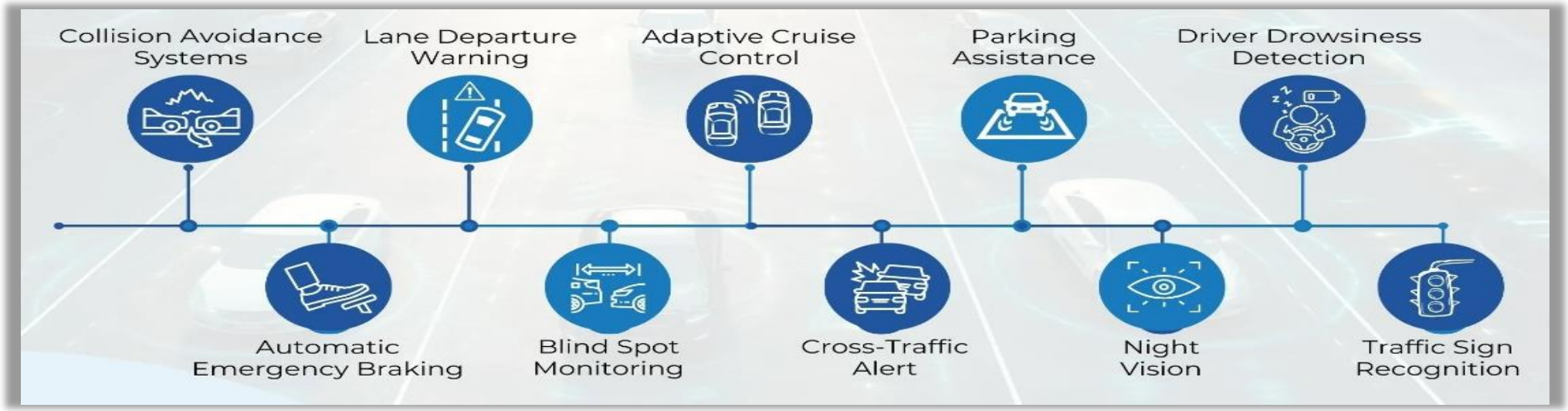

Top 15 Crash Scenarios
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: MOVING CONSTANT SPEED
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: STOPPED IN TRAFFIC LANE
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: SLOWING OR STOPPING
Substance: NONE DETECTED, Distracted: LOOKED BUT DID NOT SEE
Movement: MAKING LEFT TURN
Substance: NONE DETECTED, Distracted: LOOKED BUT DID NOT SEE
Movement: MOVING CONSTANT SPEED
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: MAKING LEFT TURN
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: ACCELERATING
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: PARKED
Substance: NONE DETECTED, Distracted: INATTENTIVE OR LOST IN THOUGHT
Movement: MOVING CONSTANT SPEED
Substance: NONE DETECTED, Distracted: NOT DISTRACTED
Movement: STARTING FROM LANE
Substance: NONE DETECTED, Distracted: LOOKED BUT DID NOT SEE
Movement: ACCELERATING
Scenario
0
5000
10000
15000
20000
25000
Crash Count
Driver At Fault
NO
YES


Driver Fault Trends by Collision Type
Driver At Fault
Number of Crashes
0
2000
4000
6000
8000
STRAIGHT MOVEMENT ANGLE
SAME DIR REAR END
HEAD ON LEFT TURN
SAME DIRECTION SIDESWIPE
SINGLE VEHICLE
ANGLE
HEAD ON
SAME DIRECTION LEFT TURN
SAME DIRECTION RIGHT TURN
FRONT TO REAR
Collision Type


Top 15 Fault Rates Across Collision Types
SAME DIRECTION RIGHT TURN
FRONT TO FRONT
SIDESWIPE, SAME DIRECTION
STRAIGHT MOVEMENT ANGLE
SIDESWIPE, OPPOSITE DIRECTION
SAME DIR BOTH LEFT TURN
SAME DIRECTION SIDESWIPE
OPPOSITE DIR BOTH LEFT TURN
HEAD ON
OPPOSITE DIRECTION SIDESWIPE
FRONT TO REAR
SAME DIR REND RIGHT TURN
SAME DIR REAR END
SAME DIR REND LEFT TURN
SINGLE VEHICLE
Fault Rate (%)
0
10
20
30
40
50
60
70
Collision Type

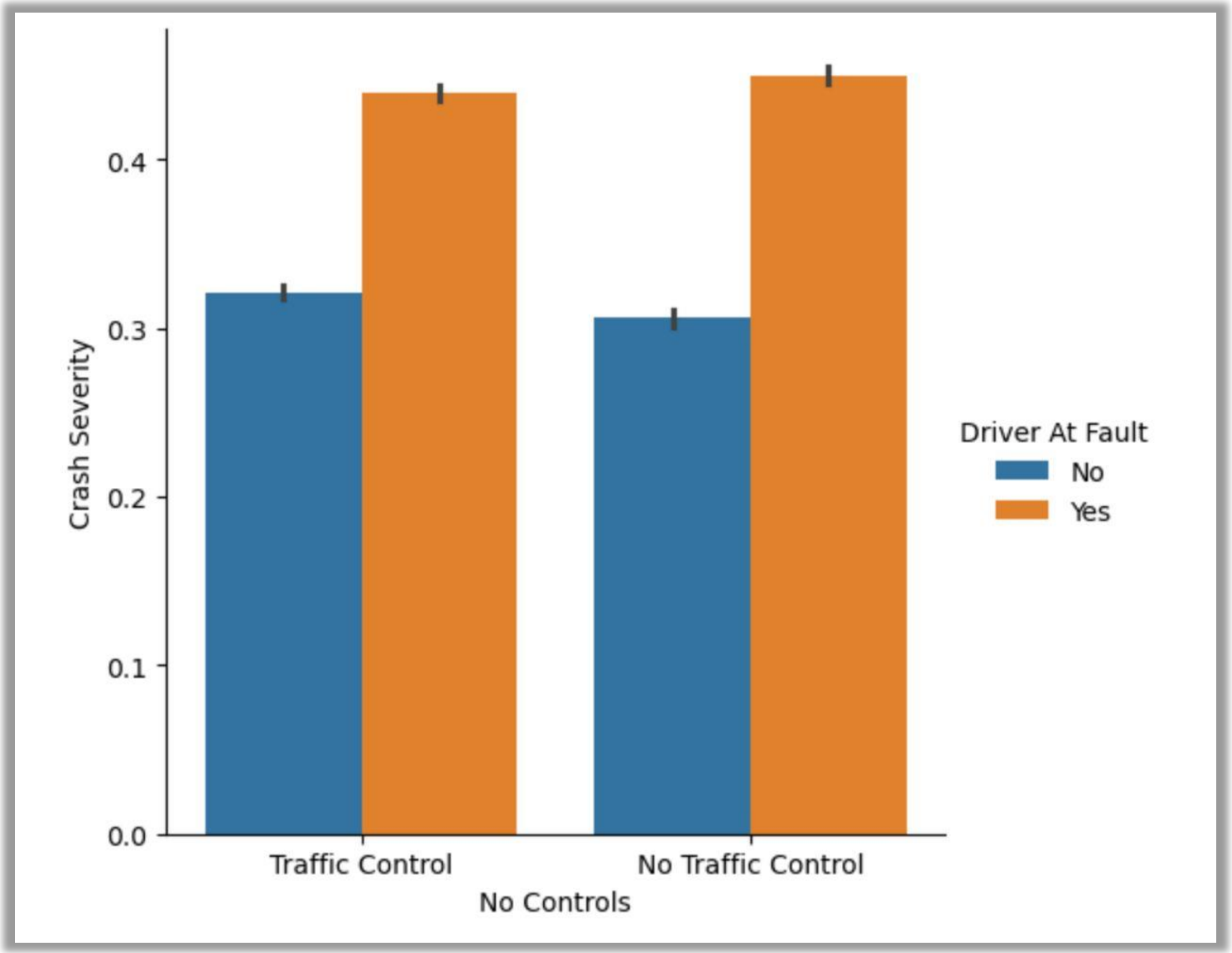
Managerial Insights

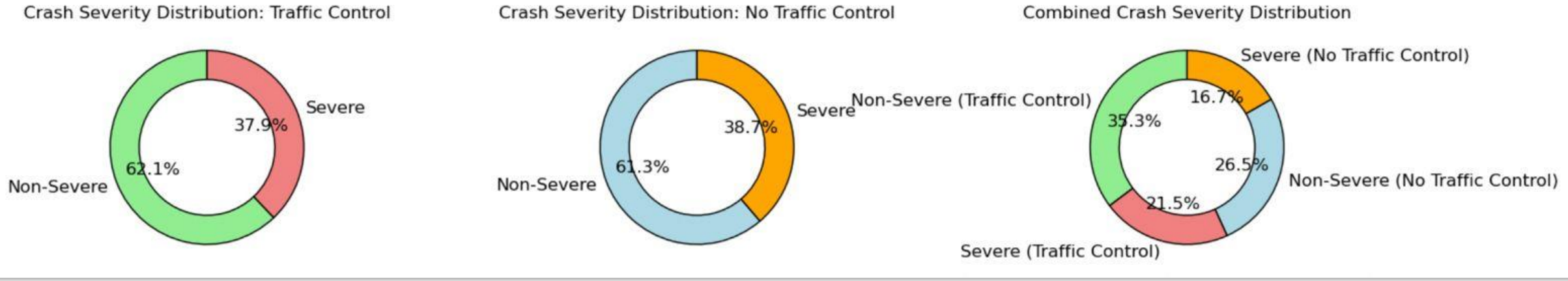
Encourage the adoption of technologies like lane-keeping assist, adaptive cruise control, and automatic emergency braking by offering tax incentives or subsidies for vehicles equipped with ADAS (Advanced Driver Assistance Systems) features.



Finding #3: Traffic control effectiveness

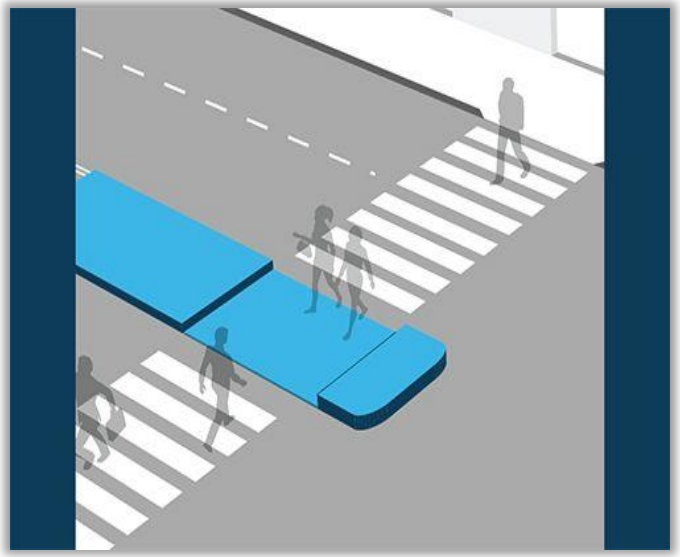
The observation that accidents and severe accidents occur at nearly equal rates regardless of traffic controls suggests that the presence of controls alone is insufficient to mitigate accident risks.


Crash Severity
0.0
0.1
0.2
0.3
0.4
Traffic Control
No Traffic Control
No Controls
Driver At Fault
No
Yes


Crash Severity Distribution: Traffic Control
Severe
37.9%
Non-Severe
62.1%
Crash Severity Distribution: No Traffic Control
Severe
38.7%
Non-Severe
61.3%
Combined Crash Severity Distribution
Severe (No Traffic Control)
16.7%
Non-Severe (Traffic Control)
35.3%
26.5%
Non-Severe (No Traffic Control)
21.5%
Severe (Traffic Control)

Managerial Insights

Expand the use of roundabouts to replace traditional intersections, promoting slower speeds and reducing crash severity. Increase the implementation of pedestrian refuge islands and dedicated bike lanes, building on existing efforts, to better protect walkers and cyclists and minimize conflicts with vehicles.



Q/A Session

Thank You